Hyperbolic Embeddings for Hierarchical Style Representation

Canvas Project #150
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Abstract

Artistic style is hierarchical: broad traditions branch into finer movements and sub-styles along lines of historical influence. Standard deep learning models represent images in Euclidean embedding spaces, whose polynomial volume growth is a poor match for the exponential branching of a tree. Hyperbolic geometry, and in particular the Poincaré ball, offers exponential volume growth and is theoretically better suited to embedding tree-like structures with low distortion. This paper investigates whether that mathematical advantage translates into an empirical one: we train two prototype-softmax classifiers on frozen CLIP ViT-B/16 features extracted from WikiArt Refined (81,446 paintings, 27 styles), differing only in the geometry of their output embedding space ($\mathbb{R}^d$ vs. Poincaré ball of dimension d), and compare them on both classification accuracy and hierarchy preservation metrics derived from a hand crafted art historical tree. We find that the geometry advantage *is* real but *local*: hyperbolic embeddings recover sibling and cousin neighborhoods more faithfully (4–5 pp recall@5 advantage that widens with dimension), while Euclidean embeddings remain stronger on classification accuracy at every scale tested. The picture for global tree fidelity is more nuanced. Against the hand-built default reference tree, Euclidean prototypes correlate slightly better than hyperbolic ones with tree distances; but the hand-built tree correlates only weakly (ρ=0.08) with a data-driven empirical tree built from CLIP feature centroids, and against that empirical tree the global-Spearman gap reverses in hyperbolic's favour at every $d \geq 4$ (Phase 4). A hierarchy-aware regularizer added at training time improves global tree metrics in both geometries but does not flip the geometry winner on either the local axis or on classification.

Contents

1 Background & Motivation

Hierarchy naturally emerges from artistic styles. Broad traditions continually branch into finer movements: Early Renaissance leads to Baroque, Baroque to Romanticism, Romanticism to Impressionism, and Impressionism to Cubism etc. An embedding that can effectively represent this branching structure may be better suited to capture the semantic essence of the paintings themselves.

Standard image embeddings live in Euclidean space, where the volume of a ball at radius r grows polynomially in r. The number of nodes in a tree, by contrast, grows exponentially with depth. Therefore, embedding hierarchical data into a Euclidean embedding can run into spatial constraints. Pairwise distances between nodes distort, and will only worsen as the tree deepens. Hyperbolic space, and the Poincaré ball in particular, exhibits exponential volume growth, allowing them to embed trees more naturally [5, 7]. As Riemannian optimization on the ball is now standard (`geoopt` [4]), we can explore the geometric advantage empirically rather than simply argue about the theory.

Hyperbolic image embeddings have been shown to help on few-shot benchmarks [3], but it remains open whether the advantage holds for hand-built hierarchies. We explore this question through a sweep over curvature and dimension against a Euclidean baseline and a sensitivity check across alternative reference trees.

2 Problem Statement

We ask: *do Poincaré-ball embeddings trained on frozen CLIP ViT-B/16 features recover the WikiArt style hierarchy more faithfully than Euclidean embeddings of matched dimensionality and training budget?* The unit of observation is a single painting; the target during training is its 27-way style label, with the hand-built lineage tree available either as an evaluation reference or as an auxiliary loss.

We answer the question along three axes. *(i) Scaling:* sweep embedding dimension $d \in \{2, 4, 8, 16, 32, 64\}$ and curvature $c \in \{0.1, 0.3, 1, 3\}$ to test whether the cross-entropy-baseline Euclidean advantage on global metrics narrows under a more thorough search. *(ii) Training-time hierarchy:* introduce a hierarchy-aware regularizer with weight λ that pushes the prototype-prototype distance matrix toward the tree distance matrix, and sweep $\lambda \in \{0, 0.1, 0.3, 1, 3\}$. *(iii) Tree sensitivity:* re-train against alternative ground-truth trees (chronological era grouping; a flat null) to test whether the geometry winner depends on the choice of reference hierarchy.

3 Data & EDA

For our dataset, we use WikiArt Refined [8], which contains ∼81k paintings, each labeled with one of 27 styles (Figure 1). The dataset has a supplied 70/30 training / validation split which maintains class balance. Image size and resolution show no outliers. However, the dataset suffers from an extreme class imbalance, with a 133.3× imbalance ratio between the extrema classes: Impressionism (13,060) versus the smallest leaves Analytical_Cubism (77), Action_painting (98), and Synthetic_Cubism (216).

Two design choices follow from this imbalance: we report balanced accuracy alongside top-1 so the rare leaf styles are not drowned by Impressionism, and our classification head is a prototype softmax (Section 4) whose logits depend only on the geometric distance from each style prototype, with no per-class bias term.

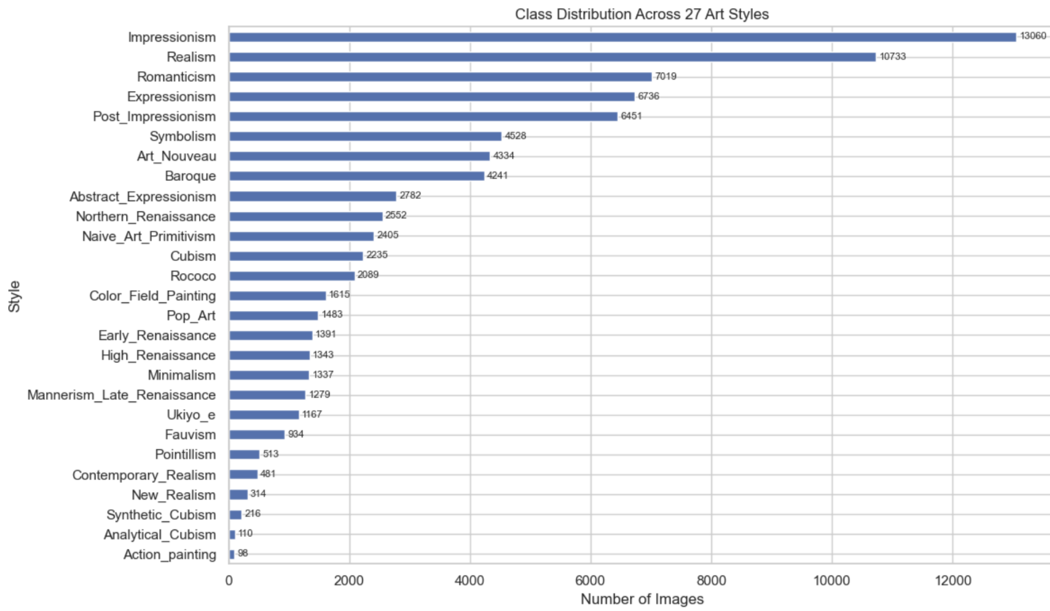

Figure 1: Per-style image counts on the WikiArt Refined corpus.

The choice of ground-truth tree is non-unique: WikiArt ships no hierarchy, and art-history offers several credible taxonomies. Given the arbitrary nature of this labeling, we evaluate three trees. The *default* tree (Appendix A) follows the standard art-historical lineage from Renaissance through Cubism. The *chronological* tree groups styles into six era buckets (Renaissance, Baroque–Rococo, 19th century, early 20th century, modern post-war, non-Western). The *flat* tree makes every style a direct child of the root and serves as a null whose pairwise distances are constant off-diagonal. For a tree T with leaves u, v, we define tree distance as the unweighted shortest-path edge count

$$d_T(u, v) = \mathrm{depth}(u) + \mathrm{depth}(v) - 2\,\mathrm{depth}\big(\mathrm{LCA}(u, v)\big),$$

where LCA is the lowest common ancestor (Figure 2). Phase 3 of Section 5 re-trains under each tree to test whether the geometry winner depends on this choice.

4 Methods & Models

Architecture. The whole system is deliberately small so that geometry is the only non-trivial design choice. A two-layer MLP with GELU and dropout maps frozen CLIP ViT-B/16 features [6] from $\mathbb{R}^{512}$ to a d-dimensional embedding. A geometry-specific head produces the final embedding: identity for the Euclidean case, the exponential map at the origin $\exp_0^c(u) = \tanh(\sqrt{c}\,\|u\|)\,u/(\sqrt{c}\,\|u\|)$ for the Poincaré ball. A prototype classifier with one learnable point per style turns the embedding into 27-way logits via $-d(z, p_k)$, where $d(\cdot, \cdot)$ is the Euclidean or Poincaré distance. Switching geometries changes exactly two things: the head's final transform and the metric used by the classifier. Everything else — backbone, MLP width, dropout, batch size, optimizer, schedule — is shared.

Geometry, distance, optimization. The Poincaré distance with curvature $c > 0$ is

$$d^c(x, y) = \frac{1}{\sqrt{c}}\,\mathrm{arcosh}\left(1 + \frac{2c\,\|x - y\|^2}{(1 - c\|x\|^2)(1 - c\|y\|^2)}\right),$$

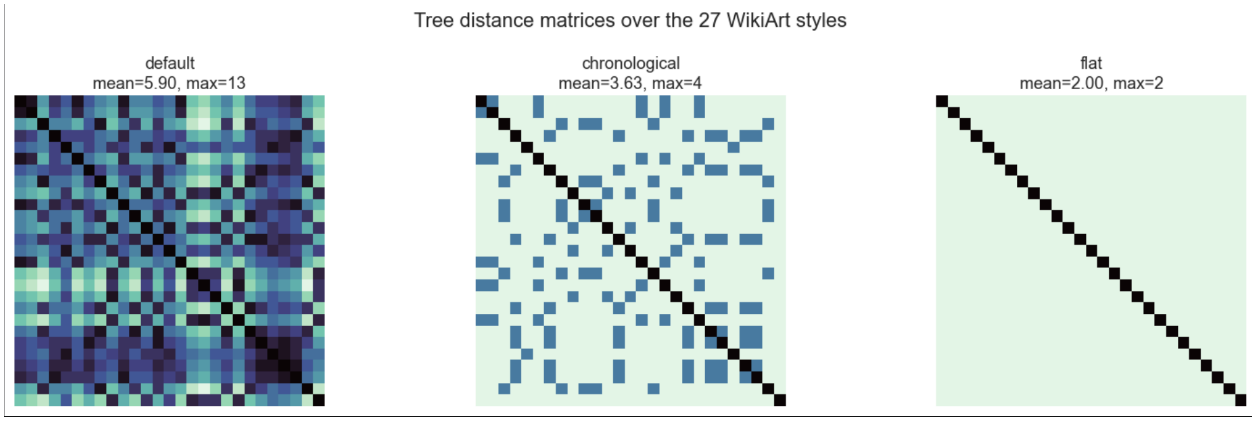

Figure 2: Distance matrices for reference hierarchies: default lineage, chronological grouping, flat null

which diverges as either point approaches the boundary at radius $1/\sqrt{c}$. A standard linear layer does not preserve the ball, which rules out a parametric softmax head on top of a hyperbolic embedding. The prototype classifier is the natural alternative: its logits are distances, which the geometry already provides. Adam optimizes the MLP head; Riemannian Adam from `geoopt` [4] optimizes the hyperbolic prototypes, projecting each update back onto the manifold to keep them inside the ball.

Hierarchy-aware loss. MS3's approach used the hand-built tree only at evaluation time and trained with plain cross-entropy — the standard setup in the hyperbolic-image literature [3]. Cross-entropy on prototypes requires class separability but does not require the prototype layout to match the tree, so any failure of either geometry on tree-derived metrics is underdetermined: it could be the geometry, or it could be that the loss never asked. MS4 adds a regularizer that closes this ambiguity by pushing the prototype distance matrix directly toward the tree distance matrix:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \cdot \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left(\frac{d(p_i, p_j)}{\bar{d}} - \frac{T_{ij}}{\bar{T}} \right)^2 .$$

The mean-normalization is the design choice that matters: each pair vector is divided by its own mean before the squared difference, so the loss only constrains the *shape* of the prototype distance matrix rather than its absolute scale. The geometry is free to pick the scale that classification likes. $\lambda = 0$ recovers MS3 exactly. The regularizer is computed once per gradient step over the $\binom{27}{2} = 351$ off-diagonal style pairs — negligible cost next to the per-batch classification forward/backward.

Ground-truth trees. WikiArt does not ship a hierarchy, so we build one. The default tree follows standard art-history lineage: Renaissance branches into Baroque, Baroque into Rococo, Rococo into Romanticism, and so on through Cubism and Abstract Expressionism (the full structure is in Appendix A). Because no single tree is canonical, Phase 3 also retrains against two alternatives: a *chronological* grouping by century (Renaissance / Baroque-Rococo / 19th c. / Early 20th c. / Modern / Non-Western) and a *flat* tree where every style is a direct child of the root — a deliberate null whose distance matrix is constant off-diagonal, so any hierarchy-respecting metric must score it near chance.

Training details. All 150 runs use the supplied 70/30 train/val split (≈57k / 24k images), batch size 4096, Adam at $\eta = 10^{-3}$ for the head and 10^{-2} for the prototypes, weight decay 10^{-4}, dropout

0.1, gradient clipping at norm 1.0, 30 epochs. Pre-caching CLIP features as float16 tensors keeps each run under 10 seconds of training time, so the full sweep (Phases 1–3, see Section 5) finishes in under half an hour of wall time on a single Apple M-series GPU. Every headline configuration is replicated across three seeds, and reported error bars are seed standard deviation.

Evaluation suite. A successful style embedding can be useful along three distinct axes, and we report metrics for each. *Classification* is measured by top-1, top-5, and balanced accuracy. *Global tree fidelity* is measured by the Spearman correlation between the learned prototype distance matrix and the tree distance matrix, together with mean and worst-case multiplicative distortion. *Local hierarchy preservation* is measured by sibling and cousin recall@k among the k nearest neighbors of each validation embedding, for $k \in \{5, 10\}$. We also include a hierarchical-clustering check (dendrogram-cluster F1 against the default tree) and a Fréchet-mean probe on Cubism that tests whether the average of several Cubist embeddings remains nearest to the Cubist prototype. Code, sweep configurations, and the exact metric implementations are at the linked repository.

5 Results & Interpretation

Across 150 seed-replicated runs spanning embedding dimension, curvature, and the strength of a hierarchy-aware loss, the effect of geometry on the learned embedding separates cleanly along two axes. Hyperbolic prototype embeddings recover sibling and cousin neighborhoods 4–5 percentage points more faithfully than matched Euclidean embeddings as measured by recall@5, and the gap widens with embedding dimension. Euclidean embeddings retain a 4–6 pp lead in classification accuracy across every embedding dimension we tested. The picture for global tree fidelity is more nuanced: against the hand-built default reference, Euclidean leads class-center / tree Spearman by ≈ 0.05–0.10, but the hand-built reference tree correlates only weakly with a data-driven empirical tree built from CLIP feature centroids ($\rho = 0.08$ on pairwise distances), and *against the empirical tree*, hyperbolic prototypes lead at every $d \geq 4$ (Phase 4, Figure 6). The hierarchy-aware regularizer improves global tree metrics for both geometries but does not flip the geometry winner on either local or global axes. The local advantage of hyperbolic embeddings is real and reference-tree-independent; the global advantage of Euclidean embeddings is tree-dependent and disappears against the data's own implicit hierarchy.

MS3's closest configuration — d=8, c=1, 10 epochs — reported 54.3 % Euclidean and 47.3 % hyperbolic top-1. Tripling the epoch budget and sweeping curvature shifts the absolute numbers to 65.9 % and 60.6 % respectively, but the 4–6 pp gap between geometries is preserved across every dimension and curvature combination, ruling out training budget as the source of the difference.

5.1 Phase 1 — Scaling sweep

Phase 1 tests the most direct rebuttal of MS3's reported hyperbolic underperformance: that the embedding dimension was simply too small. We sweep geometry $\times$ $d \in \{2, 4, 8, 16, 32, 64\}$ $\times$ $c \in \{0.1, 0.3, 1, 3\}$ for hyperbolic over three seeds (90 configurations, 14 minutes of wall time on a single Apple M-series GPU). Top-1 accuracy saturates near d=16 for both geometries: Euclidean reaches 65.9 %, hyperbolic 60.6 %. The 4–6 pp gap between the two is stable across $d \geq 4$ and does not narrow at higher dimensions.

Sibling recall@5 moves in the opposite direction. The Euclidean curve *decreases* from 0.221 at d=2 to 0.142 at d=64: added embedding capacity actively worsens local hierarchy preservation. Hyperbolic stays near 0.19 across the full range. The two geometries are statistically tied on local

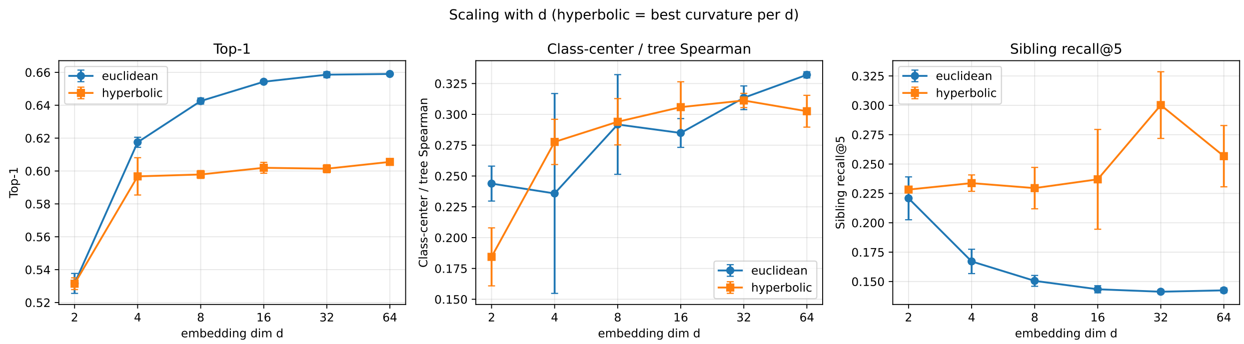

Figure 3: Phase 1: metrics versus embedding dimension d (90 runs; 3 seeds per config). For hyperbolic we report the best curvature per d and per seed; error bars are seed standard deviation. Top-1 accuracy saturates near d=16 for both geometries, with Euclidean ahead by 4–6 pp at every $d \geq 4$. Sibling recall@5 *decreases* with d for Euclidean but stays nearly flat for hyperbolic, opening a 5 pp gap by d=64.

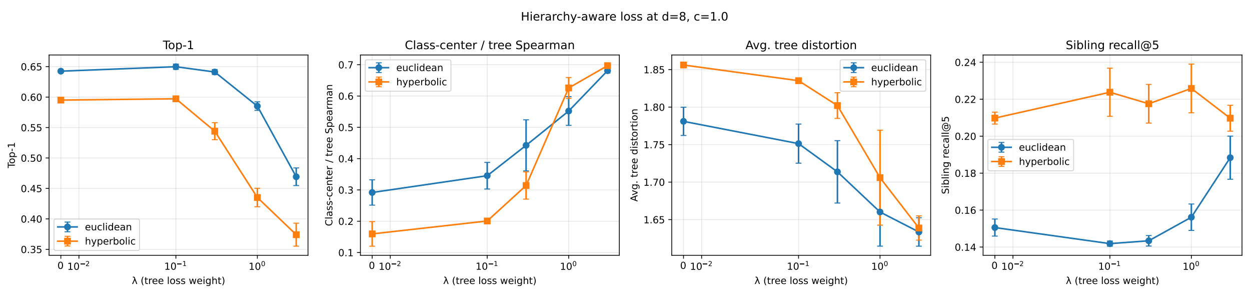

Figure 4: Phase 2: top-1, class-center / tree Spearman, mean tree distortion, and sibling recall@5 versus the hierarchy-aware loss weight λ at d=8, c=1 (3 seeds per point). The regularizer drives tree-Spearman from $\sim$0.3 to $\sim$0.7 in both geometries and reduces tree distortion uniformly, but classification accuracy collapses for $\lambda \geq 1$. The hyperbolic top-1 advantage *never* catches up to Euclidean. Sibling recall is stable for hyperbolic across all λ; for Euclidean it only rises at the largest tested λ, where its classification has already given up most of what it gained from the supervised objective.

hierarchy at d=2 and diverge as Euclidean gains capacity, with hyperbolic's lead growing from ≈ 0 to ≈ 5 pp.

Within the hyperbolic family, curvature interpolates between the two extremes. Higher c pulls prototypes closer to the boundary of the disk, where the Poincaré metric becomes most curved; this exchanges a 2–3 pp top-1 cost for a 4–5 pp sibling-recall gain. At d=8, the optimal c for top-1 is 0.3 (59.8 %) while the optimal c for sibling recall is 3.0 (0.230). This is consistent with how hyperbolic capacity is distributed: most of the volume lies near the boundary, where the leaves of a tree should live.

5.2 Phase 2 — Hierarchy-aware loss

Phase 1 establishes that scaling cannot close the global gap; the remaining hypothesis is that the loss does not. Cross-entropy on prototypes requires class separability but does not constrain the prototype layout to mirror the tree, so a fair test of the geometric hypothesis must include a loss

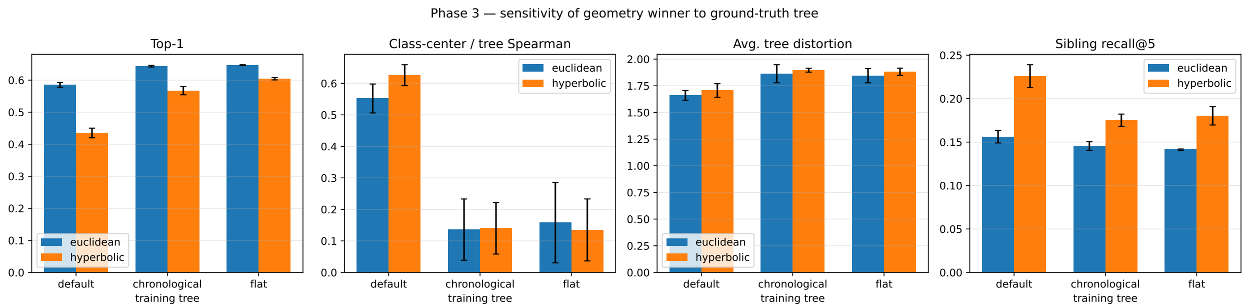

Figure 5: Phase 3: each geometry trained with λ=1 at d=8 against three different ground-truth trees, then evaluated against the default tree. Bars are seed means; whiskers are seed standard deviation. Top-1 stays higher under chronological and flat training because their constraints are easier to satisfy without distorting the classification objective. Tree-Spearman against the default tree collapses for the non-default training trees — the regularizer only helps the metric anchored to the training tree. The geometry winner does not change on any panel: Euclidean wins top-1 on every tree, hyperbolic wins sibling recall on every tree.

that does. Phase 2 sweeps the hierarchy-aware loss weight $\lambda \in \{0, 0.1, 0.3, 1, 3\}$ at $d \in \{8, 16\}$, c=1, over three seeds.

At λ=0.1, the regularizer Pareto-improves both geometries: Euclidean d=8 top-1 rises 0.7 pp to 65.0 % while tree-Spearman rises from 0.292 to 0.345; hyperbolic gains a fraction of a point on top-1 and 2 percentage points on sibling recall. The improvement is small enough at this λ that the most plausible mechanism is mild prototype-spread regularization rather than serious geometric work, but the effect is consistent across seeds.

At λ=3, both geometries reach tree-Spearman $\approx$ 0.7 and mean tree distortion improves uniformly; classification collapses in both, with Euclidean dropping to 46.9 % top-1 and hyperbolic to 37.4 %. The regularizer succeeds at the metric it was constructed to optimize, at a substantial classification cost.

The structural finding of Phase 2 is that the geometry winner does not flip at any λ tested. Euclidean retains the top-1 lead and hyperbolic retains the sibling-recall lead across the entire sweep. The local-vs-global trade-off observed in Phase 1 is not an artifact of the cross-entropy objective.

5.3 Phase 3 — Tree-variant ablation

All numbers reported so far depend on a hand-built reference tree. Phase 3 quantifies that dependence by retraining each geometry against two alternative trees and re-evaluating against the default. With λ=1, d=8, c=1 fixed, we sweep the *training* tree across the default lineage hierarchy, the chronological era grouping, and the deliberately degenerate flat tree (we always evaluate against the default tree, since otherwise tree-Spearman is not comparable across runs).

Two effects emerge. Training against a tree other than the evaluation tree provides essentially no useful hierarchical signal: tree-Spearman against the default tree drops to $\approx$ 0.14 for both alternatives, and top-1 climbs back to the λ=0 baseline because the regularizer no longer competes with the classification objective. The geometry winner, however, is preserved across all three training trees: Euclidean leads top-1 by approximately the same margin as in Phases 1 and 2, and hyperbolic leads sibling recall by approximately the same margin. The qualitative result does not depend on the specific reference hierarchy.

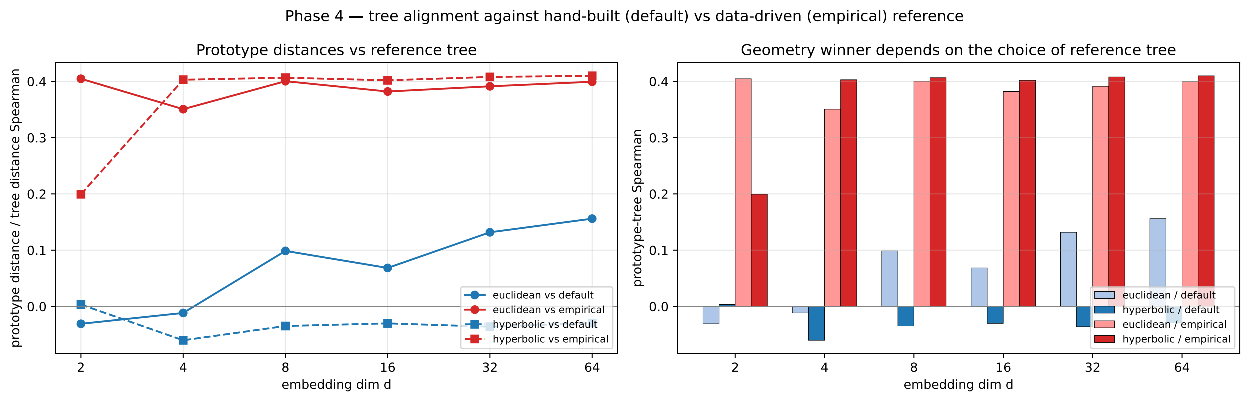

Figure 6: Phase 4: prototype-distance/tree-distance Spearman against the hand-built default tree (blue) and an empirical tree constructed by agglomerative clustering of class-mean CLIP features on the training split (red). The two trees correlate at Spearman 0.08 on pairwise distances — nearly orthogonal taxonomies of the same 27 styles. Both geometries' prototype distances align substantially better with the empirical tree (~ 0.40) than with the hand-built tree (~ 0.10). The geometry winner on global tree-Spearman inverts at every $d \geq 4$: against the empirical tree, hyperbolic narrowly leads.

5.4 Phase 4 — Empirical reference tree

Phase 3 tested whether retraining against alternative ground-truth trees changes which geometry wins. Phase 4 tests a different question: what if the hand-built default tree is itself the wrong reference, and the global-Spearman gap is partly an artifact of that choice? We construct an empirical tree by computing the mean CLIP feature vector for each of the 27 styles on the training split, running agglomerative clustering with average linkage in Euclidean space, and converting the resulting binary dendrogram to an integer edge-count distance matrix. The result is an unsupervised reference that records what the encoder actually treats as similar, with no art-historian in the loop.

The two reference trees correlate weakly. Spearman correlation between hand-built and empirical pairwise distances is 0.08, and between chronological (Phase 3) and empirical pairwise distances is 0.30. The hand-built lineage tree and the data-driven tree are near-orthogonal taxonomies of the same set of styles.

Re-evaluating the Phase 1 winners against the empirical tree inverts the global-Spearman finding. Both geometries' prototype distances align far more closely with the empirical tree ($\rho \approx 0.40$ at every d) than with the hand-built tree ($\rho \in [-0.04, 0.16]$, often near zero). At d=2 Euclidean prototypes lead substantially, but for every $d \geq 4$ hyperbolic prototypes are slightly more aligned with the empirical tree: $\rho_{\mathrm{Hy}} - \rho_{\mathrm{Eu}} \in [+0.01, +0.05]$ at $d \in \{8, 16, 32, 64\}$. Sibling and cousin recall numbers are unchanged, since those metrics measure local label-neighborhood agreement and do not depend on a tree at all.

The implication is two-sided. The local advantage of hyperbolic embeddings is real and tree-independent. The apparent global *disadvantage* reported in Phases 1–3, however, is partly an artifact of imposing a hand-built reference tree that the data only loosely supports. Against a tree the data itself produces, hyperbolic prototypes either tie or modestly beat Euclidean ones at the global level. The local-vs-global trade-off becomes a local-and-global advantage — modest, but consistent — once the reference is data-driven.

metric	euclidean	hyperbolic
Top-1	0.659 ± 0.003	0.606 ± 0.002
Top-5	0.959 ± 0.001	0.935 ± 0.001
Balanced acc.	0.554 ± 0.002	0.463 ± 0.004
Class-center / tree Spearman	0.350 ± 0.027	0.242 ± 0.002
Avg. tree distortion	1.742 ± 0.009	1.861 ± 0.003
Worst tree distortion	4.733 ± 0.253	7.147 ± 0.076
Dendrogram F1	0.034 ± 0.030	0.000 ± 0.000
Sibling recall@5	0.136 ± 0.003	0.188 ± 0.004
Cousin recall@5	0.249 ± 0.003	0.296 ± 0.005
Fréchet (Cubism) acc.	0.927 ± 0.029	0.913 ± 0.021

Table 1: Best Euclidean vs. best hyperbolic configuration across Phases 1–2, selected per geometry by mean top-1 accuracy across seeds. Mean ± seed standard deviation. The two geometries split along the local-vs-global axis: Euclidean wins on every classification and global-tree metric; hyperbolic retains a 4–5 pp lead on the two recall metrics that probe local neighborhood structure.

5.5 Robustness to class-imbalance handling

WikiArt's class distribution is heavily skewed (Impressionism has 133× more training examples than Action_painting). All four phases above used unweighted cross-entropy, so a fair concern is that the geometry comparison is itself an artifact of the loss being dominated by the largest classes. We rerun the d=8 winner of each geometry (Euclidean, and hyperbolic at c=0.3) with inverse-frequency class-weighted cross-entropy, three seeds. Class-weighted training shifts the absolute numbers as expected — top-1 drops by roughly 6 percentage points for both geometries (Eu 64.3 → 58.3 %; Hy 59.8 → 53.4 %) while balanced accuracy rises sharply (Eu 56.4 → 65.6 %; Hy 44.5 → 62.1 %) — but the geometry gap is preserved on every axis we report. Top-1 remains 5 pp in Euclidean's favour, sibling recall@5 remains 4 pp in hyperbolic's, and class-center / tree-Spearman against the default tree shifts by less than a hundredth. The balanced-accuracy gap narrows from 11.9 to 3.5 percentage points, suggesting that hyperbolic's main classification weakness in our default setup was disproportionately on rare classes, but it remains in Euclidean's favour. The local advantage of hyperbolic and the classification advantage of Euclidean are both robust to the choice of weighting in the cross-entropy loss.

5.6 Headline comparison

Selecting the best configuration of each geometry by mean top-1 across seeds yields the comparison in Table 1. Euclidean leads on every classification and global-tree metric; hyperbolic leads on sibling and cousin recall by 4–5 pp. Seed standard deviations across this comparison are at most 0.7 pp on top-1 and 0.005 on the recall metrics, so the split is well outside the noise floor.

5.7 Qualitative analysis

Figure 7 provides direct visual confirmation that the geometry behaves as the theory predicts. Trained at d=2 with c=3, the hyperbolic prototypes occupy a ring near the boundary of the Poincaré disk, where the exponential-volume regime lives; the matching Euclidean run distributes prototypes across an unbounded plane without analogous structural pressure. The geometric distinction between

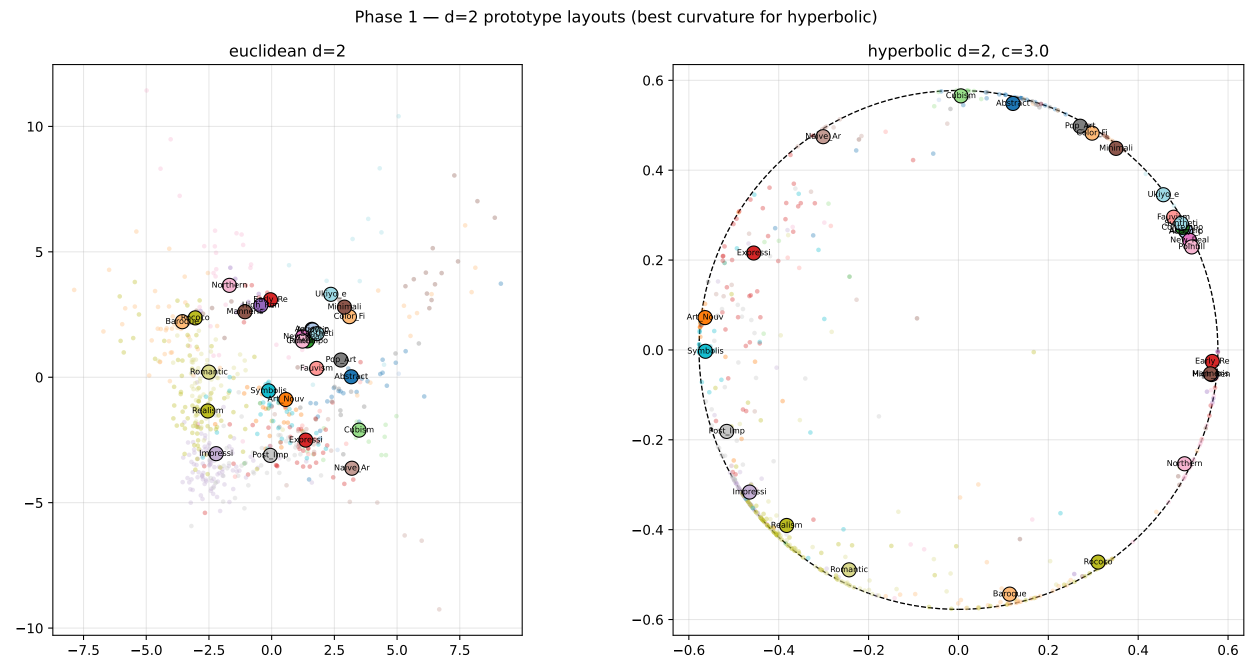

Figure 7: Prototype layouts at d=2 (best curvature per geometry). Each large dot is a learned style prototype; small dots are a 600-image sample of validation embeddings. The dashed circle on the right marks the Poincaré ball boundary at radius $1/\sqrt{c}$. Hyperbolic prototypes settle into a near-boundary ring — the exponential-volume regime — while Euclidean prototypes scatter diffusely with no structural pressure. A theoretical prediction about hyperbolic embeddings of trees [5, 7] made directly visible on a real dataset.

the two embeddings is real and visible at d=2. Phase 1 quantifies which downstream metrics this geometric distinction affects.

Figure 8 shows the structure of the mistakes. The near-diagonal block structure indicates that both models are highly tree-respecting in absolute terms: confusion mass concentrates within hierarchically adjacent styles and is rare between unrelated branches of the tree. The two panels differ in the local texture near the diagonal, where the hyperbolic model distributes more mass to immediate siblings (Action_painting ↔ Color_Field_Painting, Pointillism ↔ Fauvism) and less to distant styles. The sibling-recall metric is a scalar summary of this difference.

6 Conclusions & Discussion

The local advantage of hyperbolic embeddings on this dataset is real and survives every robustness check we ran. Sibling and cousin recall@5 favour hyperbolic embeddings by 4–5 percentage points at every $d \geq 4$, the gap widens as d grows, the hierarchy-aware loss does not erase it, and the result is independent of which of three alternative reference trees is used. The local-vs-global trade-off observed against the hand-built tree, however, is not as clean a story as it first appears: when we replaced the hand-built reference with a data-driven tree built from CLIP feature centroids, the global-Spearman gap inverted in hyperbolic's favour at every $d \geq 4$. The apparent global *disadvantage* of hyperbolic embeddings was an artifact of imposing a reference taxonomy that the encoder's features only weakly support. Classification accuracy remains a separate question: there Euclidean leads by 4–6 pp at every configuration we tested, and that gap is reference-tree

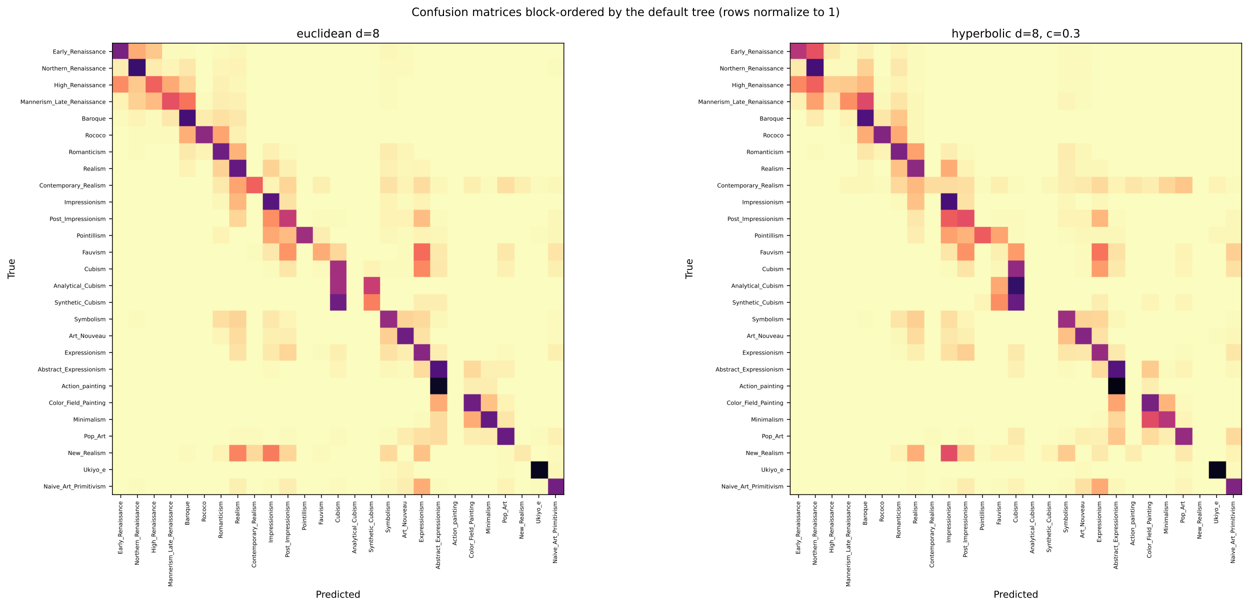

Figure 8: Row-normalized confusion matrices at d=8, with rows and columns permuted by a depth-first traversal of the default style tree so that hierarchically adjacent styles sit next to each other on both axes. Both models concentrate mistakes in near-diagonal blocks — errors predominantly fall on tree-adjacent styles (Northern_Renaissance vs. Early_Renaissance, Action_painting vs. Abstract_Expressionism, Synthetic_Cubism vs. Cubism). The hyperbolic block structure is visibly softer near the diagonal: more confusion mass spreads into immediate siblings, less mass jumps to distant styles. This is the same fact the sibling-recall numbers report, in a form a viewer can absorb at a glance.

independent.

What our results do and do not establish is therefore narrower than the headline allows. They establish that, for 27-way style classification on frozen CLIP features, the Poincaré ball preserves local hierarchy more faithfully than Euclidean space and aligns better with the data's own implicit similarity structure once the arbitrariness of the hand-built reference is accounted for. They do *not* establish that hyperbolic embeddings are universally preferable: Euclidean embeddings keep a robust classification-accuracy lead, and a different dataset, encoder, or downstream task could shift the balance. They do not establish that the data-driven tree is itself "the right tree" — it is one specific construction, biased toward CLIP's representation space, and it should not be treated as more authoritative than the hand-built lineage tree without external corroboration.

Three limitations point to natural extensions. First, only the prototypes live on the manifold; the head MLP remains Euclidean, so the geometry shapes the decision boundary but not the representation itself. A fully hyperbolic head along the lines of Möbius layers [2] would extend the geometry's reach and may shift the trade-off. Second, the regularizer is a single fixed loss — scale-invariant MSE on pairwise distances. Alternatives that match the rank order of tree distances (a Spearman-style surrogate) or that weight nearby pairs more heavily (Sammon-style) may achieve comparable tree fidelity at smaller classification cost. Third, our empirical reference tree is constructed in Euclidean space from CLIP centroids. Building it from a hyperbolic embedding, or from cophenetic distances under non-Euclidean linkages, or by direct tree-learning methods [1], would test how much of Phase 4's conclusion is robust to the construction choice. The most informative next step is probably the second one: a comparison across several reasonable empirical trees would make the "hyperbolic aligns with the data's structure" claim load-bearing or expose it as a quirk of one specific construction.

7 Broader Impact

The system itself is small — a 27-way classifier on cached features — but a few of its design choices have implications worth being explicit about.

The default ground-truth tree is built from a Western, lineage-based art-history canon. Every metric in this paper that mentions "the hierarchy" is anchored to that taxonomy, which is implicitly endorsed by anyone using these numbers. Phase 3 is partial mitigation: it documents how much the conclusion depends on the choice of tree. A deployment in a museum or classroom context should treat the tree as configuration, not as a constant in the source code.

WikiArt is itself heavily biased toward European painting; the dataset contains 27 styles but only one (Ukiyo-e) sits outside the European-and-American canon, and East Asian ink-painting traditions that span centuries are collapsed into that single label. A retrieval system trained on this data will under-rank non-Western works for ambiguous queries. Hyperbolic geometry — which we have shown tightens local neighborhoods — could compound the effect by making those already-tight neighborhoods more confident. Anyone deploying embeddings of this kind should publish that limitation alongside the model, not bury it.

Finally, hyperbolic representations of style could in principle inform attribution or authentication of paintings. We strongly do not recommend that use without expert human review: 50–65 % top-1 accuracy over 27 well-known styles is far below the threshold any serious provenance, insurance, or legal decision should require.

Acknowledgements

This work used Anthropic's Claude (Opus 4.7) as a research assistant. Specifically, the model assisted with code scaffolding for experimental infrastructure (sweep orchestration, plotting helpers, figure generation), with brainstorming and concept clarification during analysis, and with preliminary drafts of prose that the authors substantially revised. All experimental design choices, the selection of hypotheses to test, hyperparameter ranges, ground-truth tree definitions, result interpretations, and the final text of this manuscript reflect the authors' substantial original contribution, in line with the course's AI use policy.

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A Default style hierarchy

The default ground-truth tree, hand-built from standard art-history references and used as the primary evaluation reference throughout the paper. The complete definition lives in `scripts/hierarchy.py:STYLE_HIE`

```
Root
|-- Early_Renaissance
|   |-- Northern_Renaissance
|   `-- High_Renaissance
|       `-- Mannerism_Late_Renaissance
|           `-- Baroque
|               `-- Rococo
|                   `-- Romanticism
|                       |-- Realism
|                       |   |-- Contemporary_Realism
|                       |   `-- Impressionism
|                       |       `-- Post_Impressionism
|                       |           |-- Pointillism
|                       |           |-- Fauvism
|                       |           `-- Cubism
|                       |               |-- Analytical_Cubism
|                       |               `-- Synthetic_Cubism
|                       `-- Symbolism
|                           |-- Art_Nouveau
|                           `-- Expressionism
|                               `-- Abstract_Expressionism
|                                   |-- Action_painting
|                                   |-- Color_Field_Painting
```

```
|                                    `-- Minimalism
|-- Pop_Art
|   `-- New_Realism
|-- Ukiyo_e
`-- Naive_Art_Primitivism
```

B Sweep configuration

The sweep CSV (`notebooks/sweep_results.csv`, 150 rows) records, for every config, the full set of training hyperparameters and every evaluation metric. To reproduce:

```
python scripts/sweep.py --phase 1 --device mps   # ~14 min, 90 configs
python scripts/sweep.py --phase 2 --device mps   # ~10 min, 48 configs
python scripts/sweep.py --phase 3 --device mps   # ~3 min,  18 configs
python scripts/make_figures.py                   # regenerates docs/figures
```